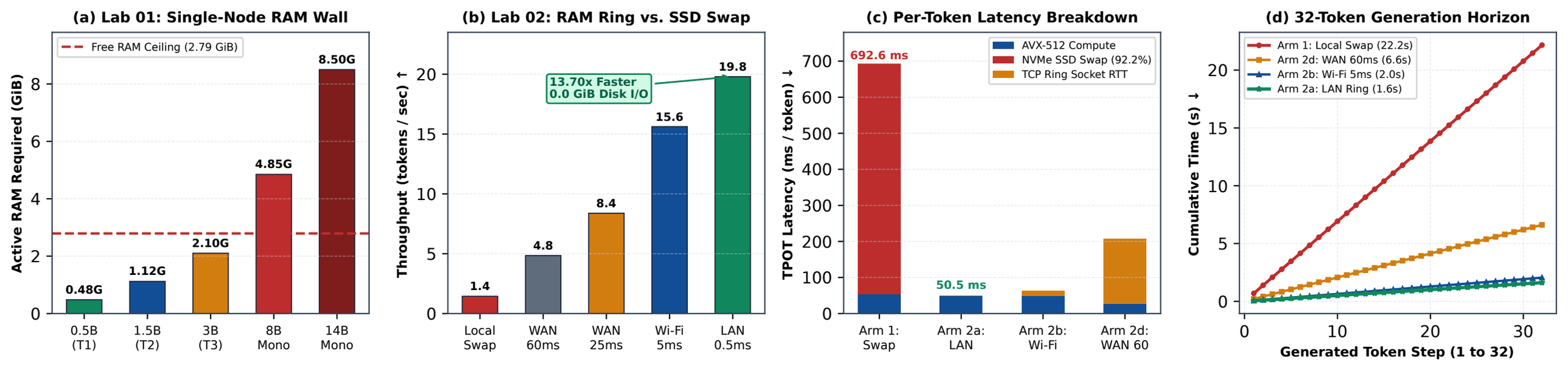
(a) Lab 01: Single-Node RAM Wall
Free RAM Ceiling (2.79 GiB)
Active RAM Required (GiB)
0.48G
1.12G
2.10G
4.85G
8.50G
0.5B (T1)
1.5B (T2)
3B (T3)
8B Mono
14B Mono
(b) Lab 02: RAM Ring vs. SSD Swap
Throughput (tokens / sec) ↑
13.70x Faster
0.0 GiB Disk I/O
1.4
4.8
8.4
15.6
19.8
Local Swap
WAN 60ms
WAN 25ms
Wi-Fi 5ms
LAN 0.5ms
(c) Per-Token Latency Breakdown
TPOT Latency (ms / token) ↓
AVX-512 Compute
NVMe SSD Swap (92.2%)
TCP Ring Socket RTT
692.6 ms
50.5 ms
Arm 1: Swap
Arm 2a: LAN
Arm 2b: Wi-Fi
Arm 2d: WAN 60
(d) 32-Token Generation Horizon
Cumulative Time (s) ↓
Arm 1: Local Swap (22.2s)
Arm 2d: WAN 60ms (6.6s)
Arm 2b: Wi-Fi 5ms (2.0s)
Arm 2a: LAN Ring (1.6s)
Generated Token Step (1 to 32)